Week 1: Fundamentals of Wireless Transmission & Architectural Differences.

1 Reviewing key structural and physical properties differentiating wired networks from wireless environments.

1.1 What differentiates the wired networks from the wireless networks?

The OSI model is used to conceptualize how data is transferred from one machine to one or more machines. The first layer of this model is the physical layer, the main purpose of which is to transmit raw bitstreams over a physical medium, using either guided media (wired) or unguided media (wireless).

The Wired Networks

The physical layer in wired networks relies on guided media to confine signals within a physical pathway. These media vary widely, including copper wiring and fiber optics.

A. **Twisted pair Cables:** In comparison to other media, twisted pair cabling is highly flexible, cost-effective, and represents the most widely implemented standard for local area network deployments.

   (a) **Physical Structure:** It consists of pairs of insulated copper conductors, typically about 1 mm thick, wrapped around each other in a helical spiral pattern resembling a DNA molecule. The specific reason for twisting these pairs is to minimize electromagnetic interference (EMI) and prevent crosstalk (noise leaking between adjacent pairs). If the wires ran perfectly parallel, the magnetic fields created by their electrical currents would disrupt the data signals.

   (b) **Classification Profiles:**

      i. **Unshielded Twisted Pair (UTP):** Consists of twisted copper pairs surrounded only by an outer plastic protective sheath. It is highly flexible and inexpensive, making it the default standard for standard office LANs.

      ii. **Shielded Twisted Pair (STP):** Encases the twisted pairs inside an internal metallic foil layer to guard against severe ambient electrical noise. It is optimized for high-interference industrial or factory environments.

B. **Coaxial Cable (COAX):** Coaxial cable offers superior shielding and higher bandwidth capacity than twisted pair lines. This allows it to span significantly longer distances at high data rates.

   (a) **Physical Structure:** It consists of a solid copper core acting as the inner conductor, surrounded by an insulating layer, which is wrapped in a cylindrical outer conductor braided mesh. The entire assembly is enclosed in a protective plastic jacket.

   (b) **Impedance Profiles:**

      i. **50-Ohm Cable:** Optimized for baseband digital transmission (historically used in early Ethernet like 10Base5/10Base2, and still used in RF antenna feeds).

      ii. **75-Ohm Cable:** Optimized for broadband analog transmission and Cable TV (CATV) systems.

C. **Fiber Opitics:** Fiber optic technology shifts the transmission medium from electrical signals over copper to light pulses over ultra-pure glass filaments.

   (a) **Physical Structure:** A core of pure glass (SiO_2) wrapped by a glass layer with a lower refractive index called cladding.

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(b) **Governing Principle:** It operates on Total Internal Reflection (TIR). When light hits the core-cladding boundary at an angle greater than the critical angle, it is trapped and guided completely within the core.

(c) **Propagation Modes:**

a. **Single-Mode Fiber (SMF):** Features a tiny core $\approx$ 8.2–10 μm. Light propagates directly down the center along a single path, eliminating modal dispersion. It uses laser sources and is deployed for long-haul backbones (spanning tens of kilometers without repeaters).

b. **Multi-Mode Fiber (MMF):** Features a wider core $\approx$ 50–62.5 μm. Light bounces down the core along multiple paths (modes). This induces modal dispersion (pulses spread out over time), capping its deployment to short-distance LANs $\leq$ 550 m.

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1.2 Structural & Physical Differences (Wired vs. Wireless):

Table 1: THE PHYSICAL ENVIRONMENT

GUIDED (WIRED) MEDIA	UNGUIDED (WIRELESS) MEDIA
• Bound physical channels	• Boundaryless, open space
• Predictable linear attenuation	• Complex, spherical propagation
• Point-to-point isolation	• Shared medium interference
• Time-invariant characteristics	• Highly time-variant channels

Table 2: Direct Structural Comparisons (Wired vs. Wireless)

Technical Property	Guided Media (Wired)	Unguided Media (Wireless)
Boundary Mechanics	Signals are physically bound within a cable sheath.	Signals propagate spherically outward through open space.
Medium Isolation	High isolation. Nearby cables do not mix signals if shielded.	Zero isolation. All devices share the exact same airspace spectrum.
Attenuation Predictability	Constant and linear over distance (logarithmic drop-off).	Highly variable, non-linear, and environment-dependent.
Channel Stability	**Time-Invariant:** Channel conditions do not change day-to-day.	**Time-Variant:** Channel changes by the millisecond due to movement.

1.3 Electromagnetic (EM) Wave Propagation Mechanics

Wireless transmitters convert electrical currents into Electromagnetic (EM) Waves that radiate into space via an antenna interface.

1. **Free-Space Propagation**: In an ideal vacuum, a transmitter connected to an Isotropic Source (a theoretical antenna radiating equally in all directions) produces a spherical wavefront. As the sphere expands outward, the energy per unit area drops because the total power is spread across a surface area that grows by $4\pi d^2$.

2. **Real-World Terrestrial Interactions (The Big Four)** When these waves travel across a real-world setting—like the Kirkuk University campus—they interact with physical obstacles via four core mechanisms:

(a) **Reflection:** Occurs when an EM wave encounters an object whose dimensions are much larger than the signal wavelength (λ), such as building walls, flat ground, or large signs. The wave bounces off the surface.

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(b) **Refraction:** Occurs when the wave passes through media with different densities (e.g., changing atmospheric layers or glass windows), causing the wave to bend.

(c) **Diffraction:** Occurs when the wave path is blocked by an impenetrable object with sharp edges, like the corners of campus buildings or hilltops. The wave bends around the edges, creating a secondary source of waves that illuminates the ”shadow region” behind the obstacle.

(d) **Scattering:** Occurs when the wave hits an object with dimensions smaller than or equal to the wavelength, or a rough surface with dense obstacles (e.g., foliage, trees, rain, or moving crowds). The wave energy is dispersed into a multitude of weaker directional rays.

1.4 Signal Attenuation & Free-Space Path Loss (FSPL)

Path loss
is the difference between the transmitted power and the received power, representing the attenuation of the signal as it travels through space.
The Friis Transmission Equation
In an ideal free-space environment, the received power (P_r) is governed by **the Friis Transmission Equation:**

$$P_r = \frac{P_t \cdot G_t \cdot G_r \cdot \lambda^2}{(4\pi)^2 \cdot d^2} = \frac{P_t \cdot G_t \cdot G_r \cdot c^2}{(4\pi)^2 \cdot d^2 \cdot f^2}$$

Where:

P_t = Transmitted power

G_t, G_r = Transmitter and Receiver antenna gains

λ = Wavelength ($\lambda = c/f$, where c is the speed of light)

d = Distance between antennas

f = Frequency of operation

Critical Mathematical Realization: The received power is inversely proportional to both the square of the distance (d^2) and the square of the frequency (f^2). Therefore, higher frequency bands (like 5G millimeter-wave at 28 GHz) experience drastically higher path loss over distance than lower frequency bands (like 4G LTE at 1.8 GHz or Wi-Fi at 2.4 GHz).

Log-Distance Path Loss Model Real environments do not behave like a vacuum. To model real paths, engineers adapt the equation into the Log-Distance Path Loss Model, expressed in decibels (dB):

$$PL(d) = PL(d_0) + 10n \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma$$

Where:

d_0 = A close-in reference distance

n = The Path Loss Exponent. In free space, $n = 2$. In dense urban zones or indoor office environments, obstructions push n to values between 3 and 5.

X_σ = A zero-mean Gaussian random variable representing Shadowing (large-scale signal blockages).

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1.5 Multipath Fading: The Urban Channel Dilemma

Because signals bounce off objects via reflection and scattering, a single transmitted signal splits into multiple copies that arrive at the receiver at slightly different times, from different angles, and with different phase shifts. This is known as Multipath Propagation.

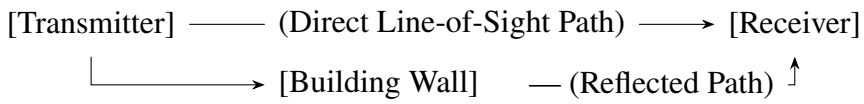


Figure 1: Multipath Propagation Diagram

These arriving waves combine destructively or constructively at the receiver's antenna. If a user moves their phone by just a few centimeters (half a wavelength), the waves can shift out of phase, causing a dramatic drop in signal strength.

Fading Scales

1. **Large-Scale Fading (Shadowing):** Slow variations in signal strength over long distances as the receiver moves behind massive obstructions like hills or buildings.

2. **Small-Scale Fading (Multipath Interference):** Rapid, violent fluctuations in signal power occurring over fractions of a wavelength or milliseconds of movement.

Statistical Distributions for Small-Scale Fading

When designing wireless receivers, engineers use statistical distributions to model these random small-scale variations:

1. **Rayleigh Fading Distribution:** Used to model environments with no clear direct Line-of-Sight (NLOS) component. The received signal is composed entirely of indirect reflected and scattered waves. This represents dense downtown centers or cluttered indoor rooms.

2. **Rician Fading Distribution:** Used when there is a dominant, direct Line-of-Sight (LOS) path alongside weaker multipath reflections. The stronger the direct line-of-sight component, the more stable the channel becomes.

1.6 Basic Antenna Principles

An antenna is a specialized transducer that converts bound electrical currents traveling through transmission lines into unguided EM waves radiating through space, and vice-versa.

Key Performance Metrics

1. **Isotropic Radiator:** A purely theoretical antenna that radiates power perfectly and uniformly in all directions. It has a gain of 1 (or (0, dBi)) and serves as the mathematical baseline for measuring real antenna performance.

2. **Antenna Gain (G):** A measure of the antenna's ability to direct its radiated power in a specific direction relative to an isotropic source. It does not amplify power; it shapes the energy distribution.

3. **Beamwidth:** The angular separation between the points where the radiated power drops to half of its maximum value (the (3dB) points) on the antenna's radiation pattern.

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4. **Effective Aperture A_e:** The real physical metric describing an antenna's ability to capture energy from an incoming EM wave, mathematically tied to its gain via:

$$G = \frac{4\pi A_e}{\lambda^2}$$

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5. **Polarization:** The spatial orientation of the electric field (E-field) vector of the emitted EM wave over time (e.g., Vertical, Horizontal, or Circular). For optimal signal recovery, the transmitting and receiving antennas must share the same polarization alignment.

Practical Antenna Configurations

1. **Dipole Antenna:** A center-fed antenna consisting of two straight collinear conductors, typically cut to a total length of a half-wavelength ($\lambda/2$). It creates a toroidal, doughnut-shaped omnidirectional radiation pattern.

2. **Monopole Antenna:** A single vertical conductor mounted above a conductive ground plane, acting as a quarter-wavelength ($\lambda/4$)element. This is the classic whip antenna pattern found on older routers and car radios.

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